

Fine-tuning Large Language Model for Gene Prioritization in m9.2 Erythroid Module

By

Dr. Ganesh B Regular
Associate Professor
Department Of IT

PRESENTED BY

BURUGU YASHWANTH - 22881A12D6
KATTA SIDHARTH REDDY- 22881A12F5
GOLLA POOJA CHAMUNDI-22881A12E8
NALLAMALLI SAKETH-21881A12H1

Introduction

Brief Background of the Topic:

Gene prioritization is a key step in identifying genes associated with specific diseases. Traditionally, researchers have relied on biological networks, experimental data, and domain expertise. However, with the explosion of biomedical literature and data, Large Language Models (LLMs) such as Qwen, Mistral, and Deepseek offer a powerful way to extract and analyze gene-related knowledge automatically.

Motivation:

Manual gene curation is slow and cannot keep up with the rapid pace of biomedical discoveries. There is a need to build intelligent systems that can prioritize genes based on both structured data (e.g., gene-disease networks) and unstructured data (e.g., scientific literature). LLMs can bridge this gap efficiently.

Relevance in Current Scenario:

With growing focus on precision medicine, genomics, and rare disease diagnosis, accurate gene prioritization is more crucial than ever. The integration of AI in healthcare is gaining momentum, making this research highly timely and impactful.

Real-World Applications :

- Identifying candidate genes for genetic disorders
- Supporting personalized medicine and clinical decision-making
- Assisting researchers in biomedical studies and hypothesis generation

Problem Statement

Clearly Stated Problem:

Gene prioritization is a complex and data-intensive process. Existing models often fail to effectively integrate the vast amount of biomedical literature with structured biological data, leading to suboptimal or incomplete gene ranking results.

Lacking in Existing Systems:

- Limited use of unstructured data like biomedical research papers.
- Inability to understand contextual relationships between diseases and genes.
- Lack of hybrid approaches that combine LLMs with biological network data.
- Existing models are not scalable, interpretable, or adaptable to new findings.

This Problem Needs to Be Solved:

- Accurate and fast gene prioritization can significantly speed up disease diagnosis, drug discovery, and research breakthroughs.
- Solving this challenge can reduce the time and cost of biomedical investigations.
- With the exponential growth of data, AI-enhanced, hybrid systems are essential to unlock hidden patterns and insights that traditional tools miss

Objectives

Objectives:

1. **Develop a hybrid ranking system** that combines the strengths of Large Language Models (LLMs) and structured biomedical data.
2. **Extract meaningful features** from biomedical literature using LLMs like Deepseek and Qwen.
3. **Integrate biological databases** (e.g., GeneCards, OMIM, STRING) to enhance gene ranking accuracy.

Project Aims to Achieve:

- Create a scalable, explainable, and efficient gene prioritization workflow.
- Bridge the gap between AI language models and real-world biomedical applications.
- Contribute a novel solution to biomedical informatics, aiding researchers and clinicians in identifying key disease genes faster and more accurately.

Scope of the Project

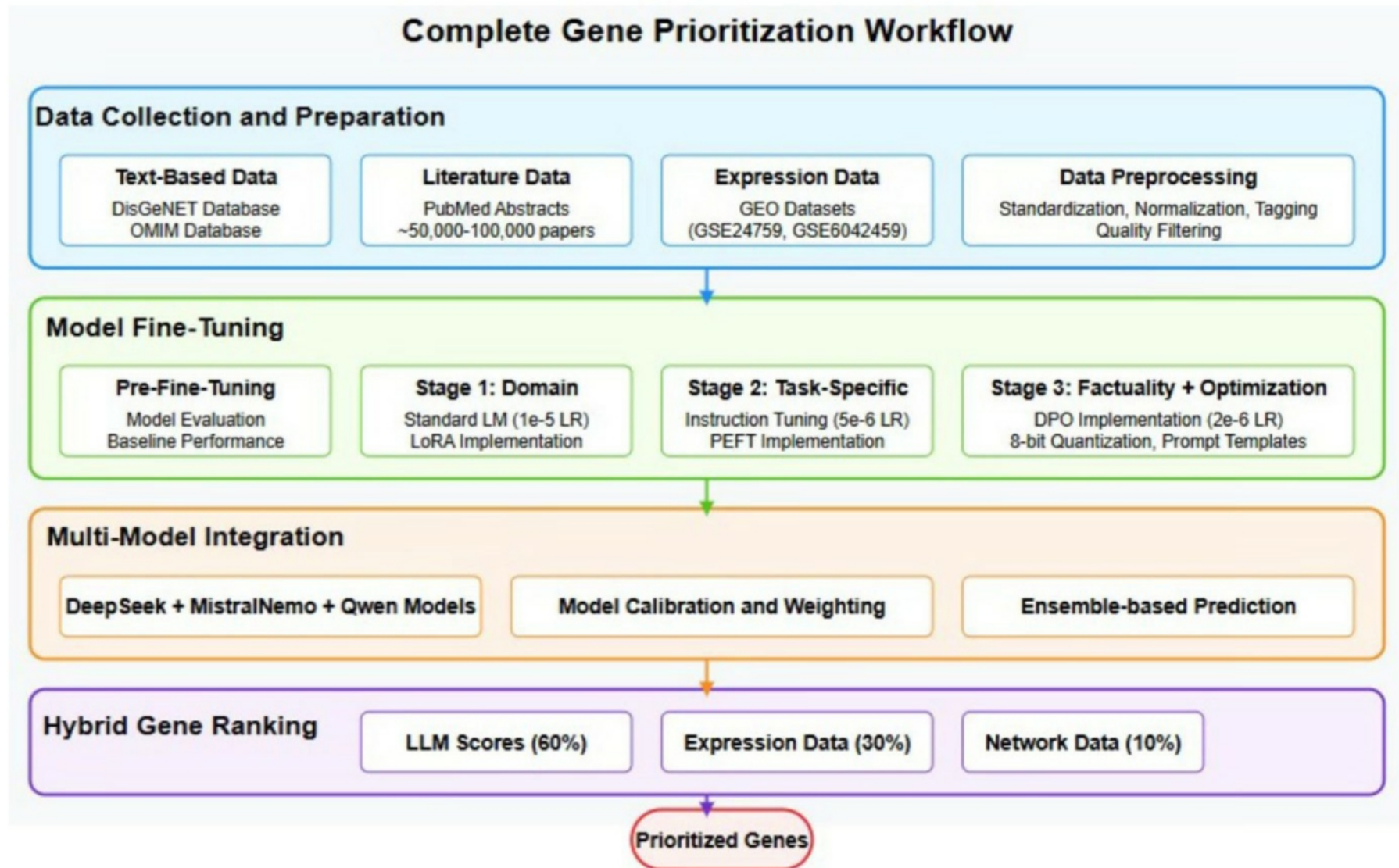
Boundaries and Limitations:

- The project focuses on human gene prioritization for known diseases using publicly available datasets.
- It uses pre-trained Large Language Models (e.g., Deepseek, Qwen and Mistral) for literature analysis.
- The study is limited to in silico (computational) analysis only — no experimental validation or wet-lab work is included.
- The models are trained and evaluated on selected benchmark gene-disease datasets.

Literature Survey Summary

Ref No.	Existing Works	Key Gaps/ Limitations	Algorithm	Our Project Addresses the Gaps
[1]	A research used machine learning (like Random Forest, SVM) to find disease-related genes based on gene networks.	They only used biological data. They didn't use any knowledge from scientific text or articles.	Random Forest, SVM	We use Large Language Models (LLMs) to understand and include information from medical literature.
[2]	Another work used BioBERT (a language model trained on medical text) to find useful information in research papers	It only found keywords/entities but didn't help in ranking genes or integrating with other methods.	BioBERT	We use Qwen not just to find information but also to improve gene ranking in our system.
[3]	A method used a hybrid of co- expression data and network info to rank genes.	They did not use any text- based or literature-based information.	PageRank, Co-expression Score	We add knowledge from research papers using LLMs to make our ranking more accurate.
[4]	A study ranked genes using Gene Ontology (GO) data and similarity scores.	It didn't work well for rare diseases or new terms not found in GO.	GO Similarity Scoring	LLMs help us understand new terms from medical articles, even if they're not in GO.

Proposed Methodology



Novelty / Innovation

Unique Features of Our Solution:

1. **Uses Large Language Models (LLMs)** like BioBERT to understand medical research papers and extract hidden gene-disease relationships.
2. **Combines two powerful approaches** — traditional biological data (like gene expression) + LLM-based text features.
3. **Creates a hybrid ranking system** that gives more accurate and meaningful results.
4. **Can work with new or rare diseases** because it learns from recent research articles, not just old datasets.

How It's Different from or Better Than Existing Methods:

- Existing methods rely only on biological or network data; they miss out on valuable text information from literature.
- Our method brings in context and meaning from biomedical articles, which improves the quality of gene ranking.
- It is more flexible and adapts to new discoveries without needing major changes to the system.
- Better accuracy and stronger predictions due to the use of AI + traditional methods.

Expected Outcomes

What do we plan to deliver?

1. A working gene prioritization model that combines AI and biological data.
2. A full project report explaining how the system works.
3. A research paper that can be submitted for academic purposes.
4. A clear workflow or diagram showing how the system works step-by-step.

The expected benefits or impact?

1. Better accuracy in finding disease-related genes.
2. Helps researchers and doctors save time by giving more relevant gene suggestions.
3. Can be used for drug discovery or personalized medicine.
4. Encourages the use of AI in bioinformatics, opening new research paths.
5. Useful for rare diseases where data is limited but recent articles are available.

Team Roles & Work Distribution

Team	Work Distribution
Burugu Yashwanth	Journal
Katta Sidharth Reddy	Code
Golla Pooja Chamundi	Base Paper
Nallamalli Saketh	Base Paper

Timeline / Gantt Chart

Phase	Duration	Tasks
Review 1	Week 1–3	Literature Survey, Problem Definition
Review 2	Week 4–6	Methodology, Tool Setup
Review 3	Week 7–9	Implementation, Testing
Review 4	Week 10–12	Results, Report, Presentation

Results

Model	Metric	Base Model (Zero-shot)	Stage 1 (Domain Adaptation)	Stage 2 (Instruction Tuning)	Stage 3 (Factually Enhancement)
DeepSeek	Scoring Accuracy	62.4%	74.8%	81.3%	85.7%
	Hallucination Rate	18.6%	14.2%	10.7%	6.3%
	Evidence Precision	58.9%	67.5%	73.8%	79.2%
	Evidence Recall	61.3%	68.9%	72.5%	76.8%
Mistral-Nemo	Scoring Accuracy	59.8%	70.2%	78.5%	83.1%
	Hallucination Rate	21.3%	16.7%	11.5%	7.9%
	Evidence Precision	56.7%	64.1%	71.2%	76.3%
	Evidence Recall	58.2%	65.9%	70.3%	74.1%
Qwen	Scoring Accuracy	60.5%	68.4%	75.9%	81.2%
	Hallucination Rate	20.1%	15.3%	12.2%	8.5%
	Evidence Precision	57.2%	63.8%	69.5%	74.1%
	Evidence Recall	59.2%	64.7%	70.1%	73.8%

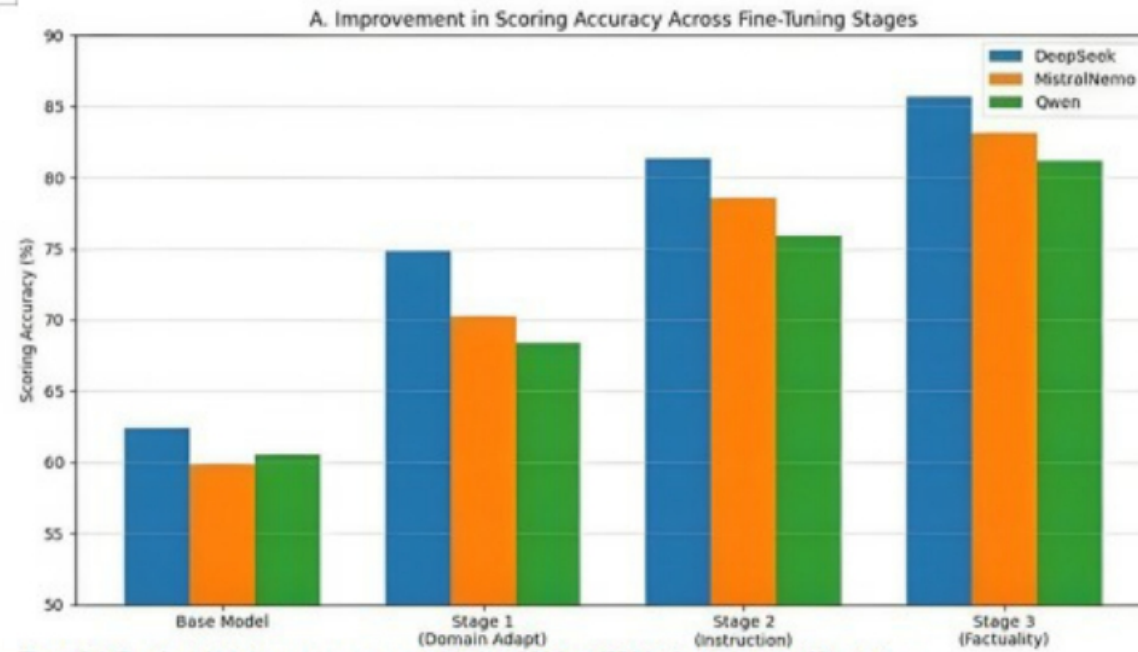
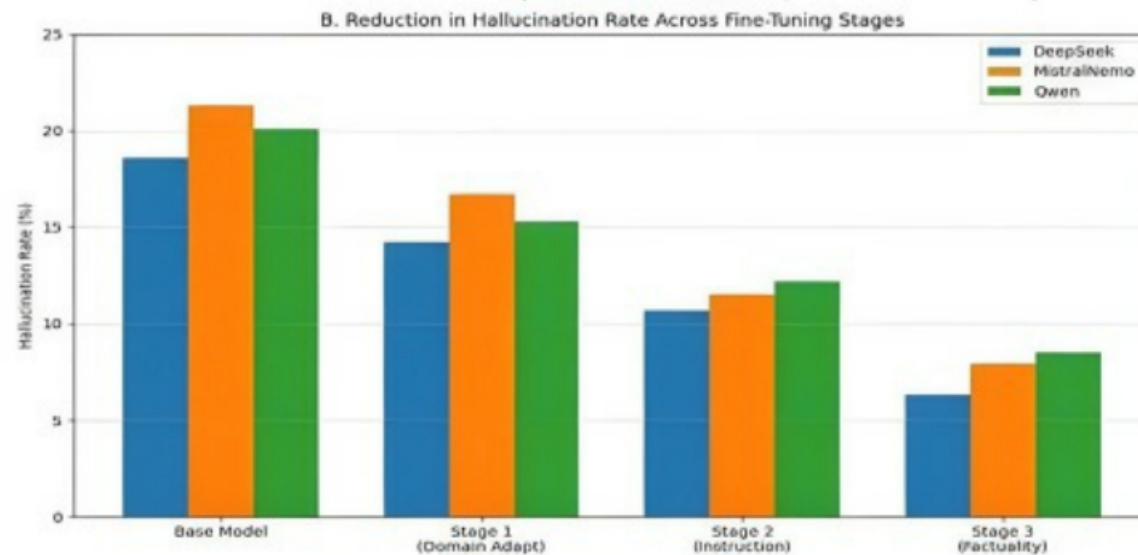


Figure 5.1: Bar chart depicting scoring accuracy of the base LLMs and LLMs after each stage of fine-tuning



References

- Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C. H., & Kang, J. (2020).
BioBERT: a pre-trained biomedical language representation model for biomedical text mining.
Bioinformatics, 36(4), 1234–1240.
- Gu, Y., Tinn, R., Cheng, H., Lucas, M., Usuyama, N., Liu, X., ... & Gao, J. (2021).
Domain-specific language model pretraining for biomedical natural language processing.
ACM Transactions on Computing for Healthcare (HEALTH), 3(1), 1–23.
- Piñero, J., Bravo, À., Queralt-Rosinach, N., Gutiérrez-Sacristán, A., Deu-Pons, J., Centeno, E., ... & Furlong, L. I. (2017).
DisGeNET: a comprehensive platform integrating information on human disease-associated genes and variants.
Nucleic Acids Research, 45(D1), D833–D839.
- Stelzer, G., Rosen, N., Plaschkes, I., Zimmerman, S., Twik, M., Fishilevich, S., ... & Lancet, D. (2016).
The GeneCards suite: From gene data mining to disease genome sequence analyses.
Current Protocols in Bioinformatics, 54(1), 1.30.1–1.30.33.
- Yang, H., Robinson, P. N., & Wang, K. (2015).
Phenolyzer: phenotype-based prioritization of candidate genes for human diseases.
Nature Methods, 12(9), 841–843.

Thank You